

WASP-11b

R-1149

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_201022 · created 2026-07-31T00:56:18+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459144.7694 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.01
Transit depth (Rp/Rs)^2	3.12 +/- 0.35 [%]
Orbital Inclination (inc)	89.88 +/- 0.56
Airmass coefficient 1 (a1)	0.7616 +/- 0.0056
Airmass coefficient 2 (a2)	0.267 +/- 0.056
Scatter in the residuals of the lightcurve fit is	0.73 %
Best Comparison Star	None
Optimal Aperture	4.92
Optimal Annulus	14.61
Transit Duration (day)	0.0518 +/- 0.0016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.01 vs 0.131 (deviation 4.24σ)
Duration fit vs published	0.0518 d vs 0.1075 d (deviation 32.78σ)
Mid-time O–C	2.2 min
Depth SNR	16.6
Residual scatter	0.73 %

Light curve

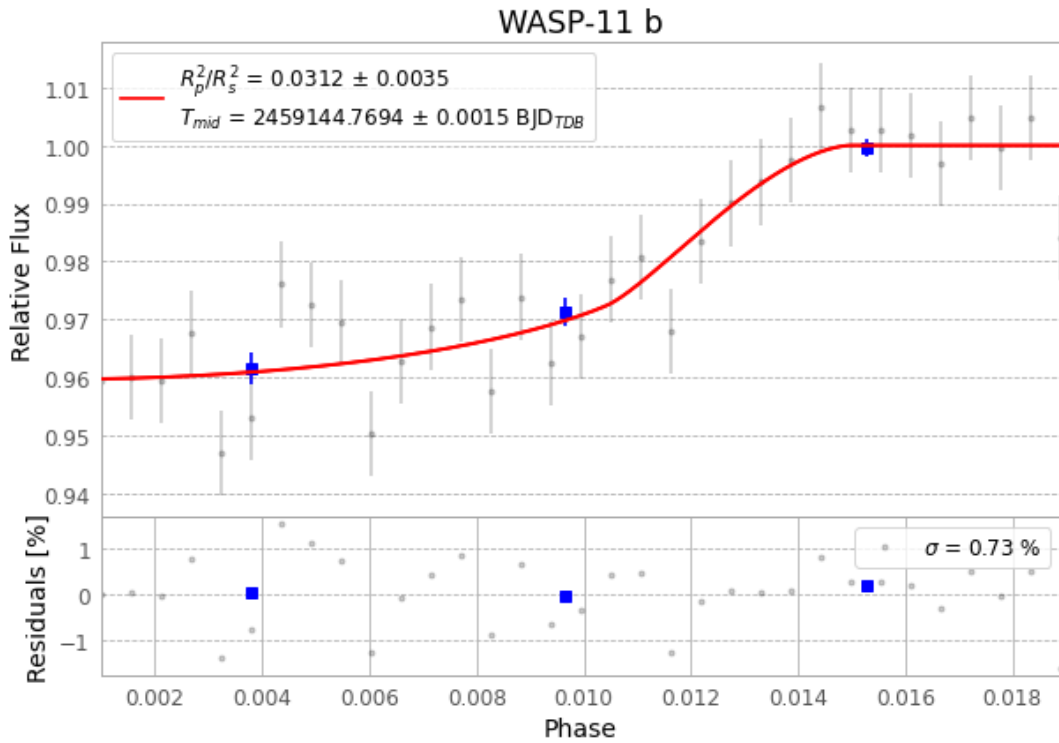


Figure 1 — FinalLightCurve_WASP-11 b_2020-10-21.png (SHA-256 2be3c32e71061d00...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [346, 312], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bfedb5e279778e76...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

33 FITS frames (22 MB), HATP-10201022062414.FITS → HATP-10201022080013.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-201022.log (c61106a6fd58...), FinalLightCurve_WASP-11 b_2020-10-21.png (2be3c32e7106...), FinalLightCurve_WASP-11 b_2020-10-21.csv (1fb5f7d823ae...)

Compute resources

Wall time 110.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 00:56Z.